

Santu Roy

AI SOLUTION ARCHITECT · AGENTIC AI SYSTEMS LEAD · ENTERPRISE ENGINEERING LEADER

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PROFESSIONAL SUMMARY

Enterprise AI Architect and Engineering Leader with **18+ years** building mission-critical platforms across global banking and fintech — in Singapore, Australia, Thailand, the USA, and India. Deep hands-on expertise in **Agentic AI systems design** : multi-LLM orchestration (Azure OpenAI, Anthropic Claude, AWS Bedrock, Google Gemini), **RAG pipeline engineering**, MCP server development, and **LangGraph-based multi-agent workflows**. Proven delivery record: **35% MTTR reduction** through agentic ITSM automation, **40% engineering effort reduction** via GenAI-powered DevOps, and 20–30% delivery optimisation across multi-cloud transformation programmes. Bridges engineering execution (Python, FastAPI, Spring Boot, Kubernetes) with enterprise architecture governance (TOGAF) and C-suite stakeholder leadership.

CORE COMPETENCIES

Agentic AI & LLMs

Azure OpenAI · Anthropic Claude · AWS Bedrock · Google Gemini · RAG (Agentic & Traditional) · MCP (Model Context Protocol) · LangGraph · Multi-Agent Systems · Prompt Engineering · NLP · LLM Evaluation

Languages

Python · Java · TypeScript · JavaScript (ES6+) · Node.js · Solidity · Shell

Back-End

FastAPI · Spring Boot · Django · NestJS · Express.js · REST APIs · GraphQL · Microservices · Kafka · Redis

Front-End

React · Next.js · Angular · Tailwind CSS · Material UI · Redux · Module Federation

Cloud & DevOps

AWS (Bedrock · EKS) · Azure (AKS · Functions) · GCP · Kubernetes · Docker · Terraform · Ansible · GitHub Actions · GitLab CI/CD · Jenkins · Azure DevOps

Data & Storage

PostgreSQL · MySQL · MongoDB · Oracle · CouchDB · Vector DBs (Pinecone · Supabase pgvector)

Enterprise & Arch

ServiceNow · ITSM · CMDB · TOGAF · Event-Driven Architecture · Microservices · API Gateway · IaC · Cloud Migration

Blockchain

Hyperledger Fabric · Ethereum · Solidity · Smart Contracts · dApps · NFTs

PROFESSIONAL EXPERIENCE

Solution Architect — AI Tech Lead

Apr 2022 – May 2026

Wipro Limited, Bangalore · Global Clients: Citibank, UBS

Spearheaded enterprise AI transformation for Tier-1 banking clients — architecting production agentic AI systems that delivered measurable reductions in operational costs and delivery cycle times at scale.

- › Architected a Unified Agentic AI Platform for intelligent ITSM operations: a multi-LLM orchestration layer spanning Azure OpenAI, Anthropic Claude, AWS Bedrock, and Google Gemini — integrated with ServiceNow across multi-cloud (AWS + Azure + GCP). Automated 60%+ of Tier-1 incident routing and reduced mean-time-to-resolution by 35%.
- › Built production MCP (Model Context Protocol) servers exposing enterprise data sources — CMDB, ITSM tickets, runbooks — as tool-callable context for LLM agents, enabling real-time grounded AI decision-making in service operations.
- › Engineered LangGraph multi-agent workflows for incident triage, root cause analysis, and automated remediation — reducing analyst investigation time and improving first-pass resolution rates in production.
- › Designed and delivered CraftBOT, a GenAI-powered infrastructure automation assistant using RAG workflows, Ansible playbook generation, and Python microservices — enabling DevOps engineers to provision infrastructure via natural language. Cut engineering effort by 40% across cloud operations.
- › Drove end-to-end cloud transformation across AWS, Azure, and GCP: IaC (Terraform, Ansible), CI/CD pipelines (GitHub Actions, GitLab, Jenkins), and observability stacks — achieving 20–30% delivery effort optimisation with measurable ROI.
- › Built production-grade microservices (Spring Boot, FastAPI, Django) and React/Next.js front-ends for enterprise-scale cloud-native banking deployments.
- › Defined transformation roadmaps, led architecture governance, and mentored cross-functional teams of 8–12 across India and offshore; owned stakeholder communication from engineering sprint to C-suite.

Solution Architect

Mar 2019 – Apr 2022

Cognizant Technology Solutions, Kolkata

- › Led enterprise conversational AI initiatives — built customer insights platforms using Rasa, Google Dialogflow, and Amazon Lex, adopted across 3 enterprise banking clients.
- › Architected the openIDL blockchain regulatory data pipeline on Hyperledger Fabric with Spring Boot, MongoDB, and AWS — enabling multi-party insurance regulatory compliance data exchange with distributed trust; significantly reduced reconciliation time.
- › Built full-stack web applications (React, Angular, Next.js) and scalable REST microservices backends (Spring Boot, FastAPI) on Kubernetes for enterprise banking clients.
- › Led hybrid cloud modernisation programmes; owned sprint governance, technical delivery, and cross-functional stakeholder management for teams of 6–10.

Solution Architect

Feb 2014 – May 2018

HP Inc., Singapore

- › Built WOobe/MOobe, HP's printing-as-a-service platform — Python, Django, Flask, REST APIs, and NLP-driven workflow automation enabling enterprise-scale print routing and SLA management for HP's global client base.
- › Designed and integrated ML models for customer behaviour analytics and predictive workflow automation across HP enterprise platforms.
- › Developed firmware validation and testing automation suites — significantly reducing manual engineering effort across hardware and software teams.

Technical Lead

Sep 2012 – Dec 2013

Standard Chartered Bank, Singapore

- › Led design and delivery of ICDMS — enterprise document and compliance management platform on Core Java with workflow orchestration for banking regulatory operations.
- › Translated complex regulatory requirements into production-grade technical architecture; drove high-performance automation frameworks for operational workflows.

Technical Manager

Jan 2012 – Jun 2012

Kasikorn Bank, Bangkok, Thailand

- › Led Temenos core banking modernisation across trade finance modules; delivered reconciliation workflows, Shell scripting automation, and MQ-based enterprise integration solutions.

Technical Lead

Apr 2006 – Dec 2011

Credit Suisse Bank, Singapore

- › Led technical delivery for GOMAN — Core Java enterprise automation for high-volume equities and derivatives trading; improved transaction workflows, reconciliation automation, and platform reliability.
- › Implemented MiFID compliance automation and MQ-based enterprise messaging for secure high-volume financial transaction processing.

KEY AI & ENGINEERING PROJECTS

Unified Agentic AI Platform for Intelligent Service Operations · Wipro / Citibank

Multi-LLM agentic system (Azure OpenAI, Anthropic Claude, Google Gemini, AWS Bedrock) integrated with ServiceNow ITSM. Built MCP servers exposing enterprise CMDB and runbooks as agent tools, LangGraph orchestration workflows, Python microservices, and a Next.js operations dashboard. Automated 60%+ Tier-1 incident routing; 35% MTTR reduction at enterprise scale.

CraftBOT — GenAI Infrastructure Automation Assistant · Wipro

RAG-powered DevOps assistant (Azure OpenAI + Python + Django + Ansible). Engineers provision cloud infrastructure, generate configuration playbooks, and execute DevOps workflows via natural language prompts. Delivered 40% engineering effort reduction across cloud operations teams.

openIDL Blockchain Regulatory Data Pipeline · Cognizant

Hyperledger Fabric blockchain pipeline for insurance regulatory reporting: Spring Boot, MongoDB, AWS. Secure distributed REST APIs and multi-party compliance data exchange workflows.

ARENA — Enterprise Engagement & Gamification Platform · Wipro

Cloud-native platform (React, Node.js, Azure Functions, MongoDB) delivering modular UI components, REST APIs, and backend automation for enterprise learning and engagement at scale.

WOobe / MOobe — HP Printing-as-a-Service · HP Inc., Singapore

Cloud-based enterprise print automation (Python, Django, Flask, NLP workflows) enabling intelligent customer routing, operational automation, and SLA management for HP's global enterprise print services.

EDUCATION

M.Sc. Information Technology
Manipal Sikkim University, Karnataka
2001 – 2004

B.Sc. Computer Science
Gauhati University, Assam
1995 – 1998

CERTIFICATIONS & CONTINUOUS LEARNING

2026 · Pursuing	
Executive Post Graduate Programme in Generative AI	IIT Kharagpur
2025	
RAG and Agentic AI — Professional Certificate	IBM
2022	
Professional Certificate in Blockchain	IIT Kanpur
2021	
Nanodegree in React	Udacity
2021	
Professional Data Science Certification	IBM